FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S HACK2SKILL

Future Health Digital Innovation Challenge

From signals to solutions.

Co-creator Microsoft

Co-creator inception42

Co-creator Mohamed bin Zayed University of Artificial Intelligence

Team Name: Nova Sentinel
Team Size: 1
Problem Statement: [Population Health] Population and Workforce Health Early Warning

Team members*
[Name, country, role, organization, age confirmation]

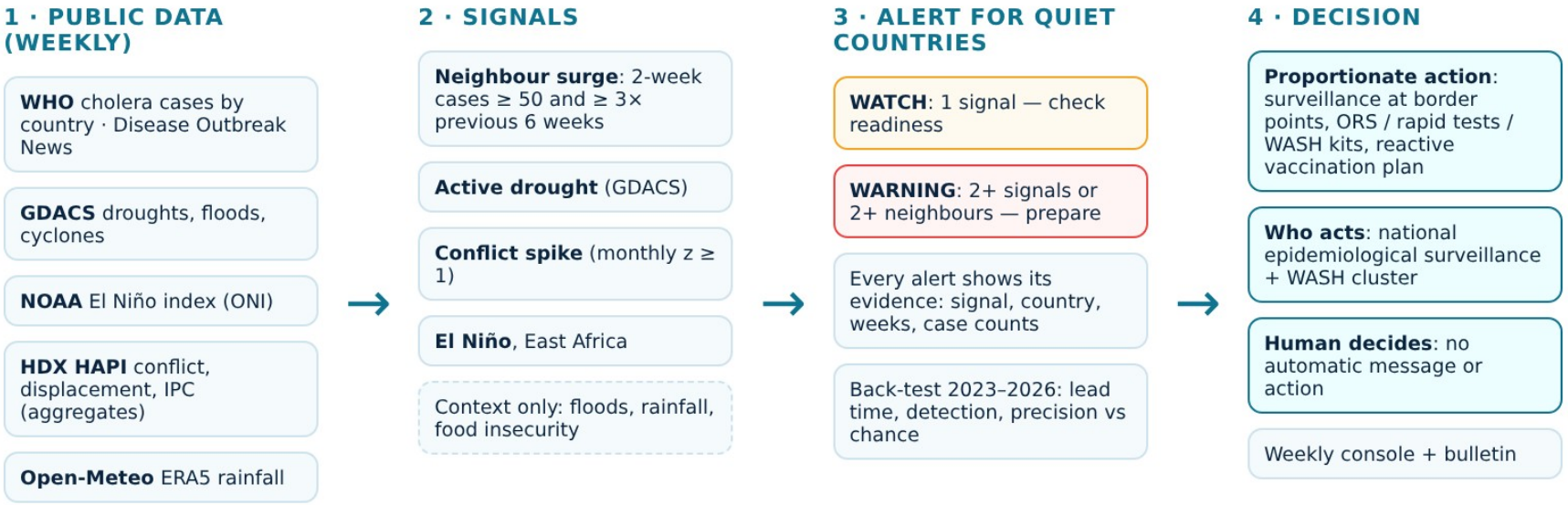
S.No	Name	Country	Role	Organization	Age confirmation
1	Ahadi Jean Cyrille Dahani	Burkina Faso	Team lead: humanitarian analyst and developer	Independent (individual entry)	18+ confirmed
2					
3					
4					
5					

Team Capability Summary*

Maximum 1,000 characters

Single-member team combining humanitarian early-warning analysis and software engineering. The lead works on context analysis, foresight and early warning for the Sahel and East Africa, with daily practice of source verification, analytical writing standards and humanitarian data (WHO, HDX, displacement and conflict aggregates). He also builds AI tooling himself: Nova, a macOS AI assistant with tool use, human-approval gates and spending controls, entered in the Meta Global AI Developer Hackathon 2026. Nova Sentinel was built during this challenge: Python ingestion of public data, transparent alert rules, a back-test on 39 cholera outbreak onsets in 26 African countries, and an interactive web console. Next step: engage a surveillance or health-cluster partner for a shadow-mode pilot.

Architecture or workflow diagram*



Target users and intended outcome*

Target users

National epidemiological surveillance units, WHO country offices, health and WASH clusters, and NGO emergency health teams in fragile and climate-exposed countries.

Intended outcome

Preparedness starts 4 to 10 weeks before cholera reaches a quiet country: surveillance of acute watery diarrhoea at border points and camps, oral rehydration salts, rapid tests and WASH kits pre-positioned, case definitions refreshed, a reactive vaccination plan ready.

Fewer late responses, lower case fatality in the first weeks, and one shared weekly evidence base for decisions. Every alert says why it fired, what to do, and who acts; a mandated officer decides.

How will success be measured?*

Measured now (back-test, Jan 2023 – Oct 2026, 39 onsets, 26 countries)

- WATCH (any one signal): 79 % of outbreak onsets preceded, median lead time 70 days.
- WARNING (2+ neighbours surging): 44 % of warnings followed by an outbreak within 10 weeks, against 20 % by chance.
- Active drought: 39 % precision, 63 days lead. El Niño (East Africa): 45 % precision.
- False-alarm episodes, data latency and behaviour with missing inputs are reported for every rule.

Pilot KPIs

- Preparedness actions started within 7 days of a WARNING · days from first case to response · case fatality in the first 4 weeks versus comparable outbreaks · alert precision in real time · user trust and alert-fatigue survey.

Evidence supporting the approach*

Citations, studies, benchmarks, or prior testing.

Our own testing

- Cross-border rule replayed on every WHO weekly series 2023–2026: 39 onsets, 19 preceded by a WATCH from the neighbour signal alone.
- Case studies: Chad 2026 (Sudan surge 4 weeks before the first Chadian cases), Chad 2025 (first confirmed case 13 July 2025, Dougui camp; WATCH 28 days earlier).
- Negative results kept: rainfall alone missed Rift Valley fever in Mauritania and Senegal 2025; rainfall plus conflict did not anticipate cholera in Ouaddaï 2025; floods and food insecurity alone were no better than chance as triggers. They are shown as context, not triggers.

Approach grounded in known drivers

- Cholera spreads along cross-border trade and displacement corridors; drought and El Niño-related extremes are recognised risk drivers. Each signal was kept only if it beat chance in our back-test.

Data sources and legal right to use them*

List source, type, owner, sensitivity, permissions, and whether personal data is used.

- WHO Global Cholera and AWD dashboard — weekly cases by country · public · WHO · aggregate, no personal data.
- WHO Disease Outbreak News API — public · WHO · no personal data.
- GDACS — drought, flood and cyclone alerts · public · EC JRC / UN OCHA · no personal data.
- NOAA Oceanic Niño Index — public domain · NOAA.
- Open-Meteo historical weather (ERA5) — CC BY 4.0 · no personal data.
- HDX Humanitarian API (HAPI) — conflict events, displacement, IPC food security · aggregated at national level · HDX terms; ACLED-derived data used only as HAPI aggregates, never redistributed raw.

Personal data: none.

All inputs are public and aggregated at country level, with free access under attribution.

Risks and safeguards*

Privacy; security; bias; explainability; misuse; human oversight; failure modes.

- Privacy: country-level data only; small-cell suppression (< 5 cases) for any future sub-national view.
- Security: read-only calls to public APIs, no credentials stored in code, runs locally.
- Bias: reporting differs between countries (testing, delays, under-reporting); uncertainty and false-alarm rates are shown next to every rule.
- Explainability: transparent rules, each alert lists its evidence (signal, country, weeks, case counts).
- Misuse: alerts serve preparedness only, are not published automatically and must not be used to stigmatise countries or communities.
- Human oversight: the system proposes a proportionate action; a mandated public-health officer decides.
- Failure modes: missing or stale feeds are flagged; 39 onsets only, so metrics carry wide uncertainty and are re-measured every week.

Hardware requirements

Required if wearable, sensor, connected device, or other hardware is there. Include sourcing and safety.

Not applicable: Nova Sentinel is software only. It runs on a standard laptop in under 3 minutes per weekly refresh, with no sensor or connected device.

If the application requires login authentication, please provide valid access credentials
or enable a "Continue as Guest" option with all necessary features and functionalities accessible to guest users.

No login is required. The prototype is a single web page opened from the link provided; all features are available without credentials. The code runs locally with Python 3, pandas and requests.

Path to pilot and scale*

Stakeholders, integration, regulatory needs, timeline assumptions, and sustainability.

- Stakeholders: national surveillance directorate, WHO country office, health and WASH clusters, NGO health partners.
- Integration: weekly bulletin and console link; later export to national dashboards (e.g. DHIS2) and sub-national data under a data-sharing agreement.
- Regulatory: no personal data; formal agreement needed only for sub-national surveillance data.
- Timeline: month 0–1 partner and data agreement · months 1–4 shadow-mode pilot over one rainy season, alerts compared with current event-based surveillance · month 5 evaluation and decision to scale.
- Sustainability: open-source code, near-zero running cost (public data, no paid model needed for alerts), maintained by partner analysts.

Pre-existing IP and third-party materials*

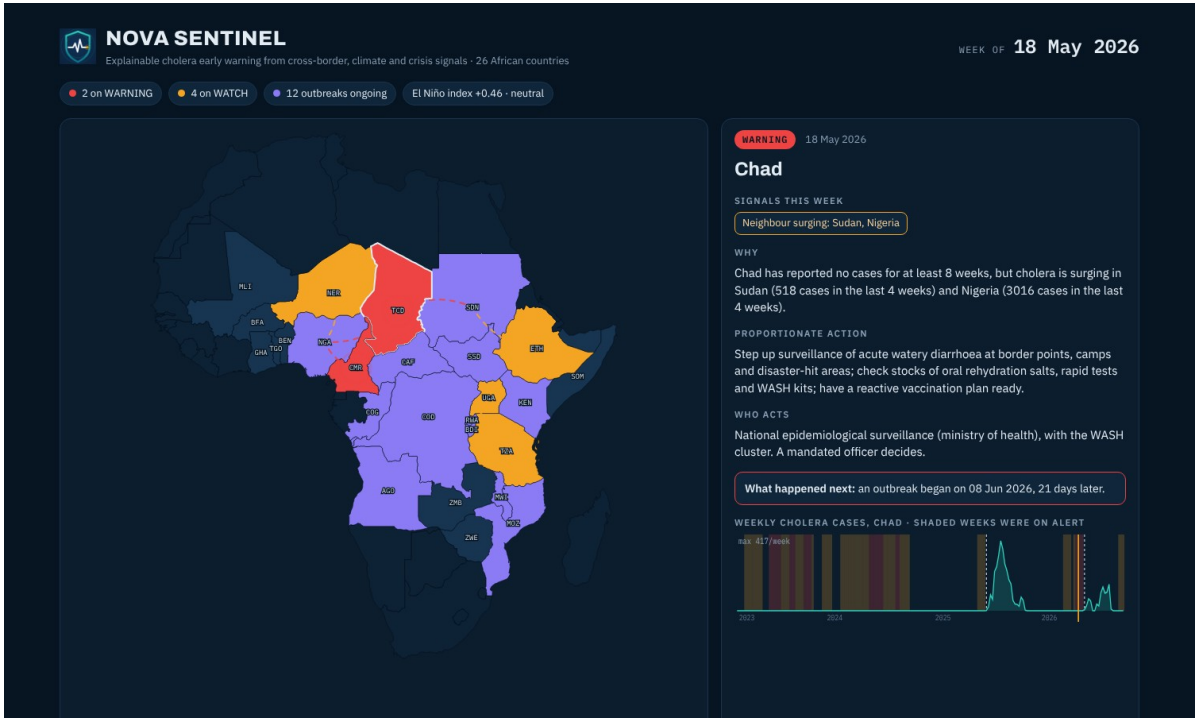
Identify ownership, licenses, open-source components, and restrictions.

- All code written by the team lead during the challenge; licence Apache 2.0.
- Open-source components: Python, pandas, requests (BSD/Apache), D3.js (ISC), world-atlas / Natural Earth boundaries (public domain).
- Data under their own licences (slide 8); no proprietary dataset, no paid model required for alerts.
- No restrictions known on use, adaptation or redistribution of the code.

Conflict of interest declaration*

Disclose relationships with organizers, partners, or judges.

None. No member of our team has a conflict of interest to declare.



Additional Slide

FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S HACK2SKILL

Future Health Digital Innovation Challenge

From signals to solutions.

Thank you

Co-creator

Microsoft

Co-creator

inception42

Co-creator

Mohamed bin Zayed University of Artificial Intelligence